

PowerFactory Assignment

Summer Semester 2026

Background

The transmission network under study has constraints on mast statics, limiting the use of larger cable cross-sections. However, high-temperature conductor cables may be considered as an optional solution. Students are tasked with performing a series of calculations and simulations to assess and enhance network reliability under different failure scenarios, while proposing and evaluating grid expansion measures.

Tasks

Students are required to complete the following tasks systematically, documenting their findings in a final report.

Task 1: Initial Network Calculations

Conduct a comprehensive analysis of the given transmission network segment using PowerFactory by performing the following calculations:

1. Load (Power) Flow Analysis:

- Execute load flow calculations to identify critical equipment (e.g., overloaded lines, transformers, or nodes).
- Document areas of concern, such as voltage violations or thermal limits.

2. $N - 1$ Failure Analysis:

- Perform an $n - 1$ contingency analysis, considering single line failures.
- Assess the network's ability to maintain stable operation post-failure.

3. $N - 2$ Failure Analysis (Common Mode Failures):

- Conduct an $n - 2$ contingency analysis, focusing exclusively on common mode failures (e.g., simultaneous failures due to shared causes).
- Evaluate the network resilience under these conditions.

4. Quasi-Dynamic Simulation:

- Simulate time-dependent network behavior under varying load and generation conditions.
- Analyze the results to identify dynamic trends or vulnerabilities.

Task 2: Enhancing $N - 1$ Security

Investigate and propose measures to achieve or maintain $n-1$ secure network operation. Potential measures include:

- Adjusting switching statuses (e.g., opening/closing circuit breakers).
- Proposing new overhead lines within the constraints of mast statics.
- Replacing existing cables with high-temperature conductors.
- **Bonus (Optional):** Exploring Optimal Power Flow (OPF) techniques to optimize network performance.

For each proposed measure:

- Create a distinct network variant or operating case in Power Factory.
- Conduct a new $n-1$ failure calculation, taking into account line failures and transient simulation.
- Evaluate whether the measure resolves identified issues (e.g., eliminates overloads or voltage violations).

Task 3: Investigating $N - 2$ Security (Common Mode Failures)

Analyze measures to ensure $n - 2$ secure operation, limited to common mode failures:

- Assess the effectiveness of your Task 2 measures under $n - 2$ conditions.
- Determine if additional adjustments are needed to mitigate risks from simultaneous failures.

Task 4: Automated Load Flow and Grid Optimization via Python API

Python API Setup and Automated Load Flow

1. Python API Setup and Automated Load Flow Connection: Write a Python script to initialize the PowerFactory application.
2. Activate your assigned project, and select the relevant operating case. Automation: Programmatically execute a standard AC Load Flow calculation.
3. Data Extraction: Extract the line loadings and bus voltages (in p.u.) for all elements in the network. Print a summary of the most heavily loaded lines to the Python console.

API Framework Hint: For optimization tasks utilizing advanced Python libraries like scipy or pyomo, running Python externally is highly recommended by running Python from an IDE like VS Code/Jupyter and calling `powerfactory.Application()`

Task 5: Optimal Active Power Curtailment / Redispatch

Using your Python environment, implement a localized optimization algorithm to address grid constraints:

The following optimization tasks based on your network state: If certain lines are overloaded during peak generation periods, formulate an optimization problem in Python to minimize the total generation curtailment (or redispatch) required to keep all line loadings strictly below 90%.

- Control Variables: Active power output (P) of controllable generators.
- Constraints: Line loading $\leq 90\%$; Generator operating limits.

Task 6: Reporting

Compile your findings into a concise report (maximum 15 pages), including:

- A clear summary of results from each task.
- Documentation of improvements achieved through proposed grid expansion measures.
- Visualizations (e.g., charts, tables, or diagrams) to present key findings effectively.
- A discussion of practical implementation challenges and limitations.
- Document your optimization methodology using tables or plots generated directly via matplotlib or in Python

Deliverables

- **PowerFactory Models:** Submit all network variants and operating cases created during the project.
- **Python script:** Submit file python script preferably as Python source code (.py).
- **Report and PFD.file:** A well-structured document (max. 15 pages) detailing your methodology, results, and insights. Along with your pfd file simulation from Power Factory
- **Submission:** Submission should be on TUWEL by the specified deadline.
- **Inquiries:** For any inquiries, send an email to *ibrahim.brahmia@tuwien.ac.at*.

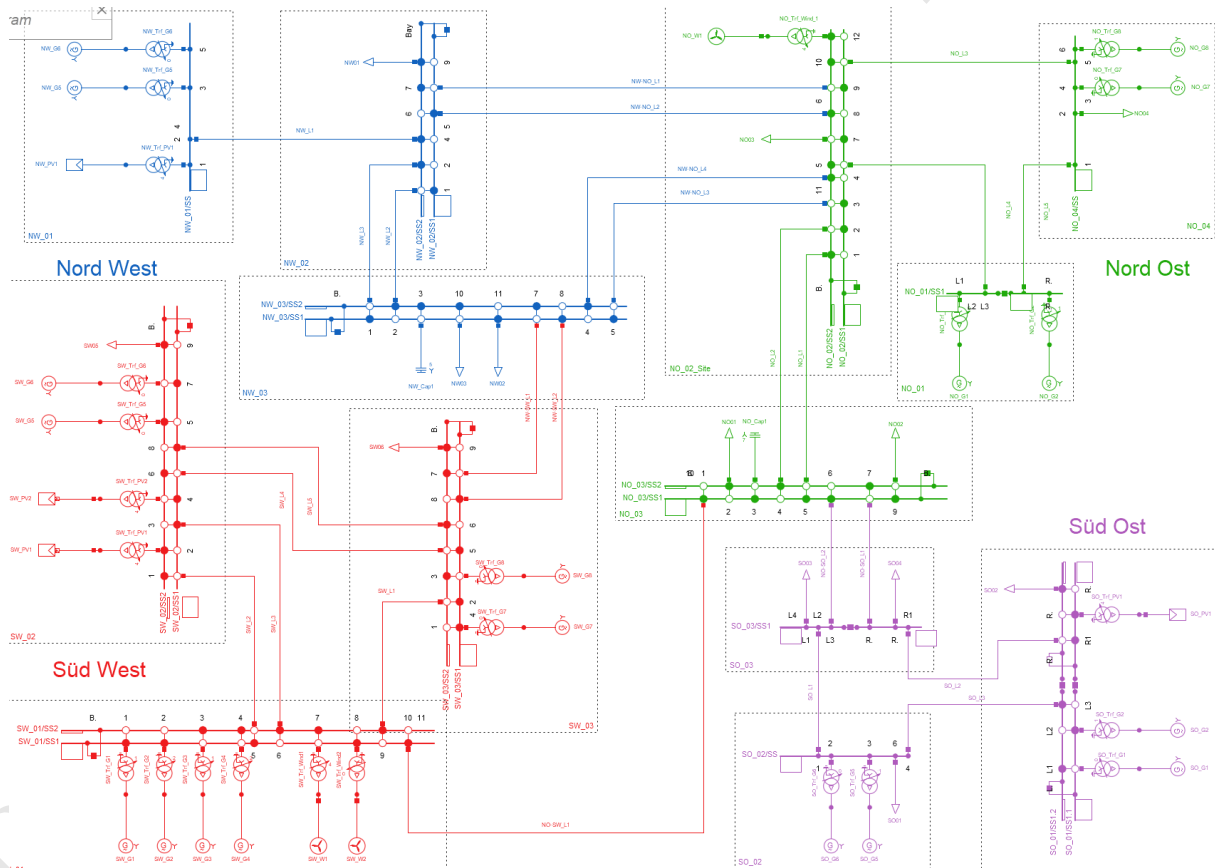


Abbildung 1: Grid Network 1

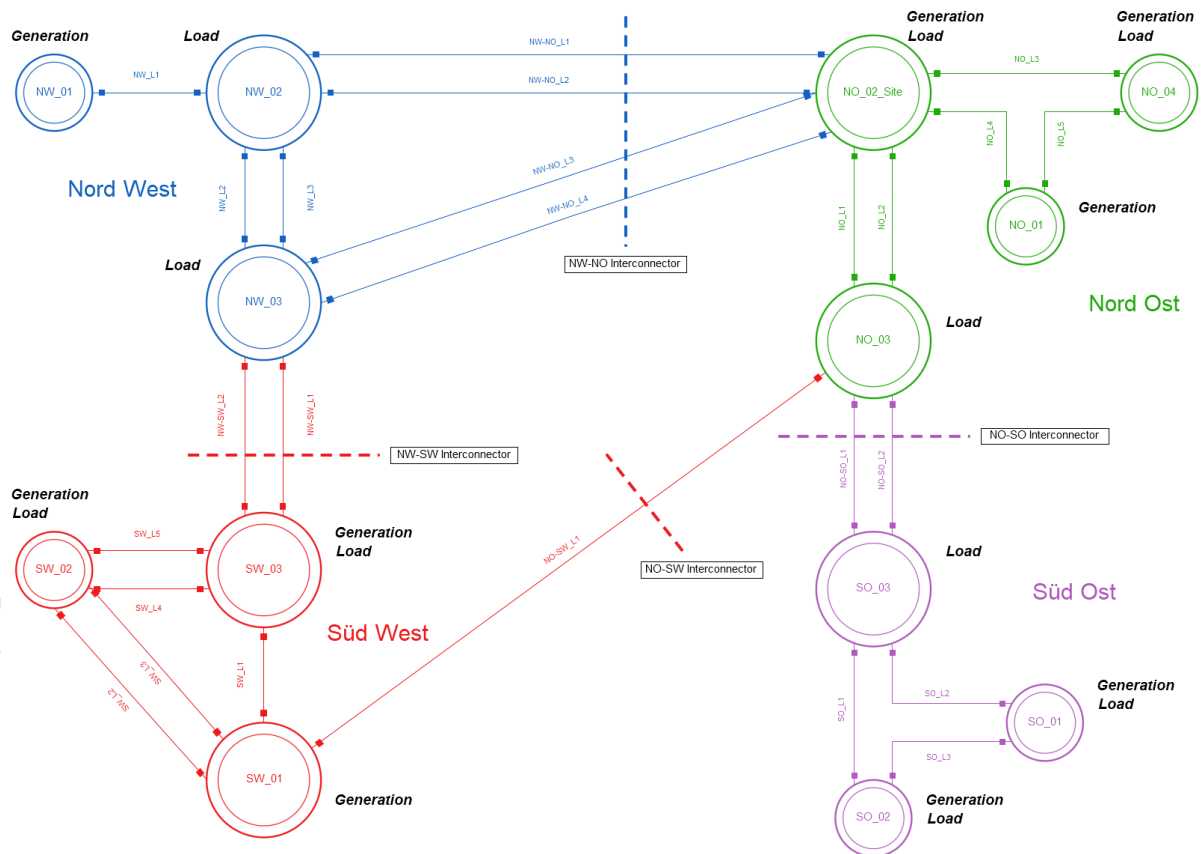


Abbildung 2: Grid Network 2